SS3 DIGITAL TECHNOLOGIES LESSON NOTES (SECOND TERM)

WEEK 1: Advanced Web Design (JavaScript & Responsive Design)

Topic: Advanced Web Design: JavaScript and Responsive Design

Introduction:
In SS1 and SS2, you learned the "Big Two" of web design:

- **HTML (HyperText Markup Language):** The **structure** or "skeleton" of a webpage.
- **CSS (Cascading Style Sheets):** The **style** or "skin/clothes" of a webpage (colours, fonts, layout).

This week, we learn the "Big Third": **JavaScript (JS)**.

- **House Analogy:** HTML is the foundation and walls. CSS is the paint and furniture. **JavaScript is the electricity, the sliding gate, and the doorbell.** It brings the house to life.

Subtopic 1.2: JavaScript (JS) Interactivity

- **What is JavaScript?** JavaScript is a high-level, **client-side** scripting language. "Client-side" means it runs in the user's web browser (like Chrome), not on a remote server.
- **What does it do?** It allows you to manipulate the HTML and CSS of a webpage after it has loaded, creating a dynamic, interactive experience.
- **Core Functions of JavaScript:**

1. **Responding to Events:** An "event" is anything the user does (e.g., `onclick`, `onmouseover`). JS "listens" for these events and runs code in response.
2. **Manipulating the DOM (Document Object Model):** The DOM is the browser's internal map of your HTML page. JS can add, remove, or change any HTML element on the page without reloading.
3. **Form Validation:** Instantly checking a user's input before they submit a form (e.g., "This email address is invalid").
4. **Making Asynchronous Requests (AJAX/Fetch):** Loading new data from a server in the background without refreshing the page (e.g., loading new tweets).
5. **Creating Sliders, Pop-ups, and Modals:** Almost any interactive component on a modern website is powered by JS.

- **Simple Code Example:**

html

```html
<!DOCTYPE html>
<html>
<head>
    <title>JavaScript Demo Page</title>
</head>
<body>
    <p id="demo">This text will change.</p>
    <button type="button" onclick="myFunction()">Click Me!</button>

    <script>
    function myFunction() {
        document.getElementById("demo").innerHTML = "Hello World!";
    }
    </script>
</body>
</html>
```

When the button is clicked, the `onclick` event triggers the `myFunction()` function, which changes the content of the paragraph.

Subtopic 1.3: Responsive Design (CSS Media Queries)

- **What is it?** Responsive Design is the practice of designing a single website that adapts and responds to the size of the user's screen. The goal is for the website to look good on any device (desktop, laptop, tablet, phone).
- **Why is it Essential?** Most internet traffic is now on mobile phones. A non-responsive website will be tiny, unreadable, and unusable on a phone.
- **The Main Technique: CSS Media Queries**
  A media query is a CSS feature that allows you to apply different styles only when certain conditions are met, like the screen's width.
- **Code Example (CSS - "Mobile-First" Design):**
  `css`

```
/* 1. Default styles (for mobile phones first) */
body {
    font-size: 16px;
    line-height: 1.5;
}
.container {
    width: 100%;  /* On mobile, use the full width */
    padding: 10px;
}
.sidebar {
    display: none; /* Hide the sidebar on mobile */
}
```

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```
/* 2. MEDIA QUERY: For tablets (screen is AT LEAST 600px wide) */
@media (min-width: 600px) {
    .container {
        width: 90%;
        margin: 0 auto; /* Center it */
    }
    .sidebar {
        display: block; /* Show the sidebar again */
    }
}

/* 3. MEDIA QUERY: For desktops (screen is AT LEAST 1024px wide) */
```

```
@media (min-width: 1024px) {
    .container {
        width: 80%;
        max-width: 1200px;
    }
    body {
        font-size: 18px;
    }
}
```

Evaluation Questions:

1. Explain the "House Analogy" for HTML, CSS, and JavaScript.
2. What does "client-side" mean in the context of JavaScript?
3. What is the DOM, and what does JavaScript do to it?
4. Define "Responsive Design."
5. What is the main CSS technique used to create a responsive design?

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WEEK 2: Advanced Database Management (SQL Basics)

Topic: Advanced Database Management: SQL Basics

Introduction:

In SS1 and SS2, we learned what a database is (tables, records, fields). Now, we learn how to **talk** to one. **SQL (Structured Query Language)** is the universal programming language for creating, retrieving, and managing data in a Relational Database Management System (RDBMS) like MySQL.

Subtopic 2.2: Categories of SQL Commands

1. **DDL (Data Definition Language):** Defines the structure of the database.

- CREATE: Creates a new database or table.

- `ALTER`: Modifies an existing table (e.g., add a column).
- `DROP`: Deletes a database or table.

2. **DML (Data Manipulation Language):** Manages the data within the tables.

- `INSERT`: Adds a new row (record).
- `UPDATE`: Modifies data in an existing row.
- `DELETE`: Deletes a row.

3. **DQL (Data Query Language):** The most common command, used to ask questions and retrieve data.

- `SELECT`: Fetches data from tables.

Subtopic 2.3: Practical SQL Queries

Let's imagine we have a table named `Students`:

Table: Students

StudentID	FirstName	LastName	Age	City
101	Tunde	Bello	17	Lagos
102	Fatima	Ahmed	16	Kano
103	Chinedu	Okafor	18	Enugu
104	Aisha	Lawal	17	Lagos

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A. SELECT: Retrieving Data (DQL)

- Get ALL columns and ALL rows: `SELECT * FROM Students;`
- Get only specific columns: `SELECT FirstName, LastName, City FROM Students;`
- Get data with a filter using `WHERE`: `SELECT * FROM Students WHERE City = 'Lagos';`
- Using `WHERE` with a number: `SELECT FirstName, LastName FROM Students WHERE Age > 16;`
- Using `WHERE` with `AND` or `OR`: `SELECT * FROM Students WHERE City = 'Lagos' AND Age < 18;`

B. INSERT: Adding New Data (DML)

- To add a new student:

```sql
INSERT INTO Students (StudentID, FirstName, LastName, Age, City)
VALUES (105, 'Emeka', 'Nwosu', 17, 'Onitsha');
```

C. UPDATE: Modifying Data (DML)

- **CRITICAL:** Always use `WHERE` with `UPDATE`, or you will change every row!
- To correct Chinedu's age:

```sql
UPDATE Students
SET Age = 19
WHERE StudentID = 103;
```

D. DELETE: Removing Data (DML)

- **CRITICAL:** Always use `WHERE` with `DELETE`!
- To delete Emeka's record:

```sql
DELETE FROM Students
WHERE StudentID = 105;
```

E. CREATE TABLE: Defining a New Table (DDL)

- This is how we would have made the table in the first place:

```sql
CREATE TABLE Students (
    StudentID INT PRIMARY KEY,
    FirstName VARCHAR(100),
    LastName VARCHAR(100),
    Age INT,
```

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```
    City VARCHAR(50)
);
```

- `INT`: Integer
- `VARCHAR(100)`: A variable-length text string (up to 100 characters).
- `PRIMARY KEY`: A constraint that says `StudentID` must be unique for every row.

Subtopic 2.4: Database Reports

A database "report" is simply the output of a well-structured query.

- `ORDER BY`: Sorts the results. `SELECT * FROM Students ORDER BY LastName ASC;`
- `COUNT`: Counts the number of rows. `SELECT COUNT(*) FROM Students WHERE City = 'Lagos';`

Evaluation Questions:

1. What does SQL stand for?
2. What are the 3 main categories of SQL commands? (DDL, DML, DQL)
3. Write the SQL command to get only the `FirstName` and `Age` of all students from the `Students` table.
4. Write the SQL command to find all information about the student whose `StudentID` is 102.
5. What is the critical keyword you must use with `UPDATE` and `DELETE`?

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WEEK 3: Networking III (Network Security & VPNs)

Topic: Networking III: Network Security and VPNs

Introduction:

A network is a "highway" for data. Network security is the practice of protecting that highway—and the "cargo" (data) on it—from thieves, spies, and vandals (hackers).

Subtopic 3.2: The "CIA" Triad of Security

The goal of all security is to protect three things:

1. **Confidentiality:** Keeping data secret and private. Only authorized users can see it. (Achieved by: Encryption, Passwords)
2. **Integrity:** Keeping data accurate and trustworthy. Only authorized users can change it. (Achieved by: Hashes, Digital Signatures)
3. **Availability:** Ensuring data and services are online and accessible when needed. (Achieved by: Firewalls, Redundancy)

Subtopic 3.3: Key Security Tools & Threats

- **A. Firewalls**
  - **What:** A network security device (hardware or software) that acts as a "gatekeeper" or "bouncer" for a network.
  - **Function:** It monitors and filters all incoming and outgoing network traffic based on a set of rules.
  - **Analogy:** A firewall is the security guard at the gate of your school. They check the ID of everyone trying to enter and can stop a student from leaving with stolen property.
- **B. Common Threats**
  - **Malware (Malicious Software):** The "umbrella" term for any software designed to cause harm.
    - **Virus:** Attaches itself to a clean file and spreads when the file is opened.
    - **Worm:** A standalone program that self-replicates across a network without human help.
    - **Ransomware:** Encrypts your files and demands a ransom payment to unlock them.
    - **Spyware:** Secretly monitors your activity to steal passwords or personal info.
  - **Phishing ("Fishing"):** A "social engineering" attack. The hacker impersonates a trusted brand (like your bank) and sends a fake email or text message. The message creates urgency and tricks you into clicking a link to a fake website where you enter your password.

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- **Denial of Service (DoS / DDoS) Attack:** An attack on Availability. The hacker uses a "botnet" (an army of infected computers) to flood a target server with fake requests, causing it to crash and become unavailable to real users.

Subtopic 3.4: VPNs (Virtual Private Networks)

- **What is a VPN?** A VPN is a technology that creates a secure, encrypted connection (a "tunnel") over a public, untrusted network (like the internet).
- **Analogy:**
  - **Without a VPN:** You are driving your car on a public road. Anyone watching can see your car and where you are going.
  - **With a VPN:** A VPN builds a private, underground, encrypted tunnel from your computer to the VPN server. Your data is scrambled before it enters the tunnel.
- **The "3 C's" - What a VPN Does:**

1. **Concealment (Privacy):** It hides your real IP address and replaces it with the IP address of the VPN server, making you anonymous.
2. **Confidentiality (Security):** It encrypts your traffic. This is CRITICAL on public Wi-Fi. Without a VPN, a hacker on that Wi-Fi can steal your passwords. With a VPN, all they see is scrambled gibberish.
3. **Circumvention (Access):** It allows you to bypass geo-restrictions, making it appear as if you are accessing the internet from another country.

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Evaluation Questions:

1. What are the three components of the "CIA" Triad of security?
2. What is the main job of a Firewall?
3. Explain the difference between a Virus and a Worm.
4. What is a Phishing attack?
5. Explain the "encrypted tunnel" analogy for a VPN.
6. List the "3 C's" (benefits) of using a VPN.

WEEK 4: Digital Project Management

Topic: Digital Project Management

Introduction:
A "project" is a temporary effort to create a unique product, service, or result (e.g., building a website). **Project Management** is the discipline of planning, organizing, and managing resources to successfully complete the project on time, within budget, and to the required quality.

Subtopic 4.2: The "Iron Triangle" (Triple Constraint)

- **The Rule:** Every project is managed by balancing three competing constraints. You cannot change one without affecting the others.

1. **Scope:** What are we building? (The features, the tasks).
2. **Time:** How long will it take? (The schedule, the deadline).
3. **Cost:** How much will it cost? (The budget, the resources).

- **The Project Manager's Job:** To balance this triangle. For example:

  - Want it faster (Time ↓)? You must either pay more (Cost ↑) or cut features (Scope ↓).
  - Want more features (Scope ↑)? You must add more time (Time ↑) or pay more (Cost ↑).
  - Want it cheaper (Cost ↓)? You must take longer (Time ↑) or cut features (Scope ↓).

Subtopic 4.3: Teamwork in Digital Projects
Digital projects require a team of specialists.

- **UI/UX Designer:** Designs how the product will look and feel.
- **Frontend Developer:** Builds the part the user sees (HTML, CSS, JS).
- **Backend Developer:** Builds the "engine" (server, database, logic).
- **QA (Quality Assurance) Engineer:** Tests the product to find bugs.
- **Project Manager (PM):** The leader. Manages the schedule, budget, and team.

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- **Key to Teamwork:**
  - **Collaboration:** Using tools that let everyone work on the same project.
  - **Communication:** Regular meetings (e.g., "Daily Stand-ups").
  - **Version Control:** A system (like Git) that tracks every single change to the code, so if someone makes a mistake, you can "rewind."

Subtopic 4.4: Project Management Tools

- **A. KANBAN BOARDS (e.g., TRELLO)**
  - **Concept:** A simple, visual way to manage tasks. It uses a "pull" system.
  - **How it works:** You have a board with columns (lists) that represent stages of a workflow.
    - **To Do** (The list of all tasks)
    - **Doing** (What the team is working on now)
    - **Done** (Completed tasks)
  - Tasks are "cards" that you move from left to right.
  - **Best for:** Agile teams, small projects.
- **B. GANTT CHARTS (e.g., MS PROJECT, ASANA)**
  - **Concept:** A complex bar chart that visualizes the entire project schedule over time.
  - **How it works:**
    - Every task is a horizontal bar.
    - The length of the bar shows how long the task will take.
    - It shows dependencies (e.g., you can't "Test" until "Build" is finished).
  - **Best for:** Waterfall projects, large complex schedules.

Evaluation Questions:

1. Define "Project Management."
2. What are the three points of the "Iron Triangle" (Triple Constraint)?
3. A client tells you, "I need the project finished in half the time, but I can't pay you more." According to the Iron Triangle, what must happen?

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4. Explain the "To Do / Doing / Done" workflow of a Kanban Board.
5. What is the main difference between a Gantt Chart and a Kanban Board?

WEEK 5: Midterm Test

Topic: Assessment of Weeks 1-4

This week is for a formal assessment of all topics covered so far this term: JavaScript/Responsive Design, SQL, Network Security/VPNs, and Project Management.

WEEK 6: Emerging Tech III (Quantum & Nano)

Topic: Emerging Technologies III: Quantum and Nanotechnology

Introduction:
This week, we look beyond today's technology to the "deep future." These are technologies, based on fundamental physics and chemistry, that will change everything.

Subtopic 6.2: Quantum Computing

- **Classical Computing (Your Laptop):** Works on **Bits**. A Bit is a simple switch that can only be a **0** (Off) **OR** a **1** (On).
- **Quantum Computing:** Works on **Qubits** (Quantum Bits). Thanks to a quantum-physics property called **"Superposition,"** a Qubit can be a **0, a 1, or BOTH at the same time.** This allows a quantum computer to perform millions of calculations simultaneously.
- **Analogy: Finding a Book in a Library**

- **Classical Computer:** You walk down every single aisle, one by one, until you find the book. (Takes a long time).
- **Quantum Computer:** You can instantly check every aisle in the entire library at the exact same moment. (Incredibly fast).

- **What will it be used for?**

1. **Breaking Encryption:** A large quantum computer could instantly break the encryption (like HTTPS, VPNs) that protects all global banking and military secrets.
2. **Drug & Materials Discovery:** Simulating complex molecules to invent new medicines or materials.
3. **Complex Simulations:** Modeling climate change or complex financial markets.

- **Current Status:** Highly experimental. Massive, super-cooled, and very unstable. A "Quantum Laptop" is far in the future.

Subtopic 6.3: Nanotechnology

- **What is it?** Technology, engineering, and science conducted at the **nanoscale**—the "atomic" or "molecular" level. A **nanometer (nm)** is one-billionth of a meter.
- A human hair is ~80,000 nm wide.
- A DNA strand is ~2.5 nm wide.
- **Analogy:** It is manufacturing and engineering with individual atoms as your building blocks.
- **What is it used for? (Applications are already here!)**

1. **Medicine (Nanomedicine):** Designing "nanobots" or "nanoshells" that can carry a drug directly to a single cancer cell and ignore the healthy cells.
2. **Materials Science:** Creating **Carbon Nanotubes**, a material 100x stronger than steel but 1/6th the weight, for use in airplanes and sports equipment. Creating nanoparticle coatings that are self-cleaning or anti-bacterial.
3. **Electronics:** Making computer chips smaller, faster, and more power-efficient.

Evaluation Questions:

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1. What is the difference between a classical "Bit" and a "Qubit"?
2. What is "Superposition"?
3. Explain the "library" analogy for Classical vs. Quantum computing.
4. What is the "nanoscale"? (How many nanometers in a meter?)
5. List two current or future applications of Nanotechnology.

WEEK 7: Midterm Break

This is a break from instruction. Students are encouraged to review the first half of the term's work and begin reviewing SS1/SS2 notes.

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WEEK 8: Revision of SSCE Past Questions I (Theory)

Topic: Revision of SSCE Past Questions: Theory

Introduction:
For the next two weeks, our goal is SSCE (WASSCE/NECO) preparation. This week, we focus on **Theory (Essay-Style) Questions**, which test your depth of knowledge and ability to explain concepts clearly.

Subtopic 8.2: Strategy for Theory Questions

1. **Read the Verb:** Pay attention to the action word.

- **"List" / "State":** Give a simple list. DO NOT write an essay.
- **"Define":** Give a clear, one-sentence definition.
- **"Explain" / "Describe" / "Discuss":** Define the term, explain it, and always use an example.

- **"Differentiate":** Give a head-to-head comparison, ideally in a table.

2. **Use Keywords:** The examiner is looking for the keywords you learned. (e.g., For "Firewall," they want "monitors," "filters," "incoming/outgoing," "rules").
3. **Be Clear, Not Long:** A clear, 3-sentence answer is better than a confused 10-sentence one.

Subtopic 8.3: Worked Examples

- **Q1. (a) What is a computer network? (b) List and explain two (2) types of networks.**

- **(a) Answer:** A computer network is a collection of two or more computers and devices, connected together by a communication medium (like cables or Wi-Fi), for the purpose of sharing data and resources.
- **(b) Answer:**

1. **LAN (Local Area Network):** This network covers a small geographical area, such as a single office or school. The devices are physically close.
2. **WAN (Wide Area Network):** This network covers a large geographical area, such as a country or the whole world. It connects multiple LANs together. The Internet is the world's largest WAN.

- **Q2. (a) Define 'Database'. (b) Explain the following terms with an example: (i) Field (ii) Record**

- **(a) Answer:** A database is a structured, organized collection of data, stored and accessed electronically, and managed by a Database Management System (DBMS).
- **(b) Answer:**

- **(i) Field:** A Field is a single column in a database table. It represents one specific piece of data for all records. *Example:* In a "Students" table, "FirstName" would be a field.
- **(ii) Record:** A Record is a single row in a database table. It represents one complete entry. *Example:* The row "101, Tunde, Bello, 17, Lagos" is one record.

- **Q3. Differentiate between a Virus and a Worm.**

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- **Answer:**

Feature	Virus	Worm
How it Spreads	Requires a host file and human action (e.g., opening the file).	Can self-replicate and spread automatically across a network.
Standalone	No, it is a piece of code attached to a program.	Yes, it is a standalone program.

Evaluation/Assignment:

Answer the following SSCE-style questions:

1. (a) What is the "Internet of Things" (IoT)? (b) Describe two (2) applications of IoT.
2. (a) What does OOP stand for? (b) Differentiate between a "Class" and an "Object."
3. (a) What is a VPN? (b) State two (2) major benefits of using a VPN on public Wi-Fi.

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WEEK 9: Revision of SSCE Past Questions II (Case Studies / Applied ICT)

Topic: Revision of SSCE Past Questions: Case Studies

Introduction:

This week, we practice the hardest type of question: the **Case Study** or **Applied ICT** question. These give you a real-world scenario and ask you to apply your knowledge to solve a problem.

Subtopic 9.2: Strategy for Case Studies

1. **Read and Identify:** Read the scenario carefully. Identify the client, the problem, and the goals.
2. **Select the Best Tool:** Choose the most appropriate technology for the problem.

- Problem: "Needs to share files in an office" -> **LAN**
- Problem: "Needs a dynamic, interactive website" -> **JavaScript**
- Problem: "Needs to store customer data" -> **Database (RDBMS)**
- Problem: "Needs to secure the office network" -> **Firewall**

3. **Justify Your Answer:** Explain *why* you chose that tool and how it solves the client's specific problem.

Subtopic 9.3: Worked Examples (Case Studies)

- **Case Study 1: The New Business**

- **Scenario:** "Mrs. Okoro is starting a new business with 15 employees in one building. She wants her staff to share files, access a single customer database, and share one printer. She also needs an internet connection, but is very worried about hackers."
- **Question:** "Advise Mrs. Okoro on the four (4) most important hardware and software components she will need to set up her office network."
- **Answer Strategy & Justification:**

1. **A LAN (Local Area Network) setup:** She will need a **Switch** and **Ethernet Cables** to physically connect all 15 computers and the printer. This creates the LAN for resource sharing.
2. **A Server:** Instead of a peer-to-peer network, she should have a central **Server**. This server will host the customer Database and shared files, making them easy to backup and manage.
3. **A Router:** This device will manage the LAN and, most importantly, provide the internet connection to all 15 staff by connecting to her ISP.
4. **A Firewall:** Because she is "worried about hackers," a **Firewall** (often included in the router) is critical. This will act as a gatekeeper, blocking unauthorized access from the internet to her private office network, thus protecting her customer database.

- **Case Study 2: The E-Commerce Website**

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- **Scenario:** "A fashion designer wants to build a new e-commerce website to sell clothes. It must look good on phones and desktops. Users must create an account and log in. The website must show live inventory. She needs to track all her sales and customers."
- **Question:** "List and describe the four (4) key technologies that will be required to build this website."
- **Answer Strategy & Justification:**

1. **HTML/CSS with Responsive Design:** HTML will structure the pages, and CSS will style them. **CSS Media Queries** must be used to ensure the site is responsive and looks good on phones, as she requested.
2. **JavaScript:** This is essential for interactive parts. It will be used for form validation (on the "create account" page) and for creating image sliders to show off the clothes.
3. **A Backend Language (e.g., Python/PHP):** She needs a server-side language to handle logic like processing payments, verifying passwords at login, and checking the live inventory.
4. **A Database (e.g., MySQL):** A database is required to store all the data, including user accounts, product details (price, inventory count), and sales records. The backend language will use SQL to query and update this database.

Evaluation/Assignment:

- **Scenario:** "The Principal of your school wants to reduce paper waste and improve communication. He wants a new system where: 1. Student results can be checked online by parents. 2. The school can send alerts to parents' phones. 3. Students can submit assignments online."
- **Question:** "Based on your knowledge, recommend and describe three (3) major digital technologies or systems that would be needed to build this 'School Portal'." (Justify why each is needed).

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WEEK 10: Mock Exam Preparation

Topic: Mock Examination Preparation

Introduction:

This week, we conduct a full-scale **Mock Examination**. This is designed to simulate the exact conditions of your SSCE. It will test your knowledge, time management, and examination strategy. The exam will cover all topics from SS1, SS2, and SS3.

Subtopic 10.2: Common Mistakes to Avoid

- **Misreading the question:** Not paying attention to verbs like "List" vs. "Explain".
- **Not using examples:** An "explain" question without an example is an incomplete answer.
- **Running out of time:** You must allocate your time. Do not spend 30 minutes on a 10-mark question.
- **Case Study Panic:** Getting intimidated by the story. Just break it down into "Problem -> Solution -> Justification" as practiced in Week 9.

Subtopic 10.3: Final Review: Top 10 "Must-Know" Concepts

1. **LAN vs WAN:** The most basic network distinction.
2. **HTML + CSS + JS:** The 3 legs of the web. What does each one do?
3. **Database (Table, Record, Field):** The structure of data.
4. **SQL (SELECT, INSERT, UPDATE, DELETE):** The 4 basic DML/DQL commands.
5. **Virus vs Worm:** The classic malware question.
6. **Firewall:** The "gatekeeper" concept.
7. **VPN:** The "encrypted tunnel" for security and privacy.
8. **Cloud (IaaS, PaaS, SaaS):** The 3 service models.
9. **OOP (Class vs Object):** The "blueprint" vs. the "house."
10. **SDLC (Waterfall vs Agile):** The two main models for building software.

Assignment:

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1. Sit for the full Mock Examination provided by your teacher.
2. After the exam, review your marked script.
3. Identify your three weakest topics and write a 1-page summary on each one.

WEEK 11-13: Revision & Examinations

Topic: Final Revision and SSCE Examinations

This period is for final, intensive revision for your SSCE (WASSCE/NECO) examinations.

- Focus on your personal weak areas identified in the Mock Exam.
- Practice writing answers under timed conditions.
- Review all your notes from SS1, SS2, and SS3.

Good luck!

This response is AI-generated, for reference only.

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